

financial gains, while holding on to assets that are losing money), momentum effect (the tendency of stocks that performed well in the past months to continue to do well in the following period and vice versa for stocks with a poor performance).

Check Your Progress 3

- 1) List the various market anomalies observed in the market.

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- 2) What is the behavioural explanation of the equity premium puzzle?

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- 3) Explain the volatility puzzle from a behavioural perspective.

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- 4) Which of the following explains the Equity premium puzzle?

- a) Under-reaction
- b) Disposition effect
- c) Momentum effect
- d) Myopic loss aversion

29.6 IRRATIONAL INVESTOR: COGNITIVE, SOCIAL AND EMOTIONAL INFLUENCES ON AN INVESTOR

Behavioural finance replaces the idea of rational decision-makers called 'econs' with 'humans' who are influenced by cognitive, social and emotional factors. Humans often are more biased than logical while making financial decisions. Before we discuss the biases that affect human decision-making, let us discuss how humans make decisions. You must have noticed that some decisions like morning routine, going to the office etc. are automatic but some decisions like financial investments etc. require attention, focus and analysis. In real life, decision-making is a complex process and is influenced by context, cues, social norms and past experiences.

When faced with difficult situations, people use heuristics or ‘rules of thumb’ to make decisions. These rules are imperfect but functional shortcuts that help investors simplify complex situations and make quick decisions.

The concept of heuristics was first introduced by the economist and cognitive psychologist Herbert A. Simon in the 1950s. He used this approach to answer the question of how humans make decisions when the conditions for rational choice theory are not met, that is how people decide under uncertainty. He suggested that there were limitations to rational decision-making.

Later, in the 1970s, psychologists Amos Tversky and Daniel Kahneman worked on ‘Heuristics’ to understand cognitive biases. Heuristics have since been used to explain the causes of irrational investors and why people’s intuitive judgement deviates from the rule. Kahneman has discussed how heuristics can be understood in terms of substitution, i.e., when faced with a difficult problem, the one that individuals are not able to address directly, they replace it with an easier question and answer that instead. For example: Instead of addressing the question: “How much profit this stock is going to make”, the investor will substitute the question: “What do I feel about this stock?” Substitution helps to provide quick answers but may lead to systematic and predictable errors called biases. In the above example, the investor has used the affect heuristic.

29.6.1 Heuristics

Four prominent heuristics are as follows:

i) The anchoring and adjustment heuristic

According to the anchoring and adjustment heuristic, people employ a certain starting point (“the anchor”) and make adjustments until they reach an acceptable value over time. Anchoring is a cognitive bias where investors rely on facts provided before a decision or an estimation is made. The facts may be completely unrelated, but research shows that they have a significant impact on the decision.

Anchoring heuristic is used by people when they have to estimate a value with an unknown attribute. In such cases, people use some initial “anchor” or default number which is then adjusted up or down based on the information available. Approximations or final values derived by a person using heuristics are often biased because anchors can never be adjusted sufficiently. Anchoring and adjustment bias occurs when an investor gives undue prominence to anchors which are most of the times statistically irrelevant. Anchoring occurs to reduce the amount of cognitive load placed on our brains.

For example: If you are asked to estimate the value of Reliance’s stock after 4 months. Since the problem is difficult and there is uncertainty, most people will apply anchoring and adjustment heuristics to find a quick response. They would use the present value of the stock as an

anchor and will adjust up or down based on any other information that they have. This may lead to a biased answer as an investor is looking at information from a twisted lens, i.e., with the present value of the stock in mind.

Property dealers often use ‘anchors’ to start negotiations. They quote higher prices for the property to set an anchor and bargain the deal to give satisfaction to the customer who is influenced by the initial anchor.

ii) The Representativeness Heuristic

According to Kahneman and Tversky (1972), representativeness means, that in situations of uncertainty, people “evaluate the probability of an uncertain event, or sample, by the degree to which it is: (i) similar in essential properties to its parent population; and (ii) reflects the salient features of the process by which it is generated”. Often people choose options or make decisions based on representative information rather than logical and probabilistic reasoning. This heuristic causes judgment errors such as neglecting base rate (people tend to ignore general information or base rate when specific information is also provided), conjunction fallacy (people assume that certain specific conditions are more likely than general conditions) and applying the law of small numbers (exaggeration of the degree to which a small sample will resemble the population from which they are drawn).

In the stock markets, investors often see the past performance of firms as being representative of the firm’s future performance as well. Investors are also seen to overweight the recent information to the detriment of the past information. Thus, if a firm has been making huge profits for a few years, investors might assume that the company will continue to perform great in future also. Representative heuristic sometimes leads to stock market overreactions and underreactions challenging the efficient market hypothesis.

iii) The Availability Heuristics

In complex and uncertain situations, the brain tries to take a mental shortcut to ease the decision-making process. These predictable shortcuts are based on our past experiences and recent memories. The availability heuristic is one such rule of thumb or shortcut. The Availability Heuristic assess the probability of occurrence of the event based on the ease with which it comes to the mind. It is governed by the principle that “if you can think of it, it must be important.” Individuals believe that the things they can recall more easily are more common and more accurate representations of the real world. For instance, the ease with which you can recall theft in your locality will determine how will you rate the law and order situation in your locality. Though a recent crime may not be a true representation of a law-and-order situation in your locality you will overestimate the occurrence of such events because of the availability

heuristic. This may lead to a cognitive bias as a recent event may not be the best representation of reality.

Availability heuristic is widely seen in financial markets also. Recent market news or events affect the investment decision more than the fundamentals of the company. Many times, this leads to an overreaction in the market which in turn might cause crashes or bubbles. In the case of a significant event, investors overestimate the probability of the occurrence of a similar event.

One example of an Availability heuristic leading to bias is the case of 'hot hand' where it is assumed that a person having a string of successes will continue with the string and will be successful. This bias was first noticed in the game of basketball (hence, the name 'hot hand') whereby the players who score maximum baskets keep on getting maximum passes even though they may be just average players. This bias is seen in the case of unrelated events like the roll of a dice or the flipping of a coin.

In financial markets, investors also experience the bias of 'hot hand' when they are more likely to deal with traders who have recently performed exceptionally well in the market. In reality, there is no relationship between past performance and future performance.

iv) The Affect Heuristic

Affect heuristic helps individuals to assign the probabilities of occurrence of an event based on how a person feels about it. If a person feels good about it, a higher probability is assigned to good consequences and vice versa.

Affect heuristic lets the decisions affected by emotions and mood while making a decision. People tend to make more risky decisions when they are happy or in a positive state of mind.

The affect heuristic is most prominent when people do not have the resources or time to reflect.

It is a bias in which emotions — fear, anxiety, surprise, pleasure — influence and quicken a person's decision-making process.

The affect heuristic may lead to overconfidence bias in an investor. Investors may make sub-optimal decisions when emotions override logic or facts.

For example: sometimes investors invest in a company's stock because "they have a good feeling about it". Good feelings may not have anything to do with the fundamentals of the company and thus, may lead to irrational decisions.

As discussed in the previous sections, Behavioural finance postulates that investors are not completely rational, preference-consistent or unaffected by emotions. Investors are boundedly rational and influenced by cognitive biases, social factors and emotions. In this section, we will discuss cognitive, social and emotional influences on an investor.

29.6.2.1 Cognitive Influences on Investor Behaviour

i) Over-Confidence

Often it is seen in financial markets that people tend to attribute success to their ability and failures to external factors. This is called self-attribution bias. This makes them overconfident about their capability which in turn leads to poor decisions regarding stocks. Overconfidence also leads to excessive trading as people having too much confidence in themselves buy too often for which they also have to sell too often (to free up capital).

Another source of overconfidence in investors is hindsight bias. Hindsight bias makes people believe after an event that they knew the outcome of the event before it happened. For example: after a stock market crash, many investors claim that they knew it was coming. When an investor believes that he has exceptional foresight or intuition, it makes him overconfident and more likely to take uncalculated risks.

ii) Cognitive Dissonance

Festinger's theory of cognitive dissonance states that people feel internal tension and anxiety when they are subjected to conflicting beliefs. Every individual tries to reduce cognitive dissonance either by changing past values or feelings or by trying to justify the decision. This is also applicable to financial market investors who attempt to justify wrong investment or decision. Cognitive dissonance leads to irrational Behaviour of selling losers too late as they are not able to accept the fact that they made a wrong decision.

This theory may apply to investors or traders in the stock market who attempt to rationalize contradictory behaviour. Investors sell losers too late as they are not willing to accept that they made a wrong decision. They also try to avoid feelings of regret or embarrassment of reporting a loss as a result of poor investment decisions.

iii) Under-reaction and Overreaction in the Market

Investors are often biased in perceiving any new information relative to their prior beliefs. Thus, if any new information or news is against their existing beliefs, they underweigh it and initially underreact to it. In the weeks following the information, they start to adjust their investments.

Similarly, when any new information, news or announcements are similar to their existing beliefs, investors overreact to it. Both underreaction and overreaction are irrational investor Behaviour and lead to sub-optimal decision-making. This leads to stock market volatility.

iv) Disposition Effects

The disposition effect refers to the tendency to prematurely sell assets that have made financial gains while holding on to assets that are losing money. Investors sell profitable investments to make quick profits while holding on to losing investments in the hope of converting them into gains. The disposition effect reflects irrational behaviour as investors lose out on possible gains due to momentum when they sell winning assets too quickly.

Daniel Kahneman has explained the disposition effect through Prospect theory. According to the theory, investors become risk-averse after gains and choose to realise the gain rather than risk losing it. On the other hand, investors become risk-seeking after a loss and hold on to the risky asset.

v) Diversification Bias

People seek more variety when they choose multiple items for future consumption simultaneously than when they make choices sequentially, i.e., on an 'in the moment' basis. Diversification is non-optimal when people overestimate their need for diversity. When they have many options, they diversify even if it is in their best interest to invest in a single asset.

On the other hand, sometimes investors diversify insufficiently. Availability heuristics and familiarity biases lead to heavy investment towards companies from investors' home countries. There is no economic justification for the home bias but familiarity reduces the perception of risks.

vi) Mental Accounting

Mental accounting refers to a set of cognitive operations that investors use to keep track of their financial investments. Investors classify their money differently based on various subjective criteria. Investors invest in both safe and speculative assets and keep these investments in different mental accounts so that negative returns from speculative assets cannot affect the positive returns from safe assets. Thus, investors keep money that they can afford to lose for speculative purposes. Rationally, there should be no division between safety capital and money that an individual can afford to lose. This mental bias often leads people to make irrational decisions and financially counterproductive investment decisions.

i) Herding

Herding in financial markets generates speculative bubbles when traders are tracking the decisions of others rather than the fundamental value of assets.

Investors with herding instincts follow others and make similar investments rather than trusting their analysis. Herd instinct can lead to Asset bubbles, panic buying or selling and can be very detrimental to the market.

Herding is the outcome of social learning when people believe that others are better informed and thus follow the crowd. During financial speculation, it is seen that investors buy at exorbitantly high prices not because they think that the asset is worth it but because they believe that others think it is.

ii) Reputation

Herding often emerges as a reputation concern as well. Keynes postulated that it is rational and follow the crowd and herd as it helps to maintain a good reputation. People believe that it is better to fail conventionally than to succeed unconventionally

iii) Beauty contests

Financial beauty contests were first used by Keynes to describe the second-guessing that characterizes financial speculation. The beauty contest is a newspaper competition where competitors are asked to select from a series of photos which according to them will be liked by the majority of people. It is similar to iterated reasoning used in financial speculation. Speculators are interested in identifying short-term investment plans of others and not what they think will be the best investment.

Financial speculations subject asset prices to financial loops which in turn might lead to financial instability. Instability is magnified if borrowings are the source of asset funding.

iv) Success Stories and Advice from Family

Investors are also affected by success stories and advice from the family. It was found in various research studies that people tend to invest more when their neighbours and family members have made profits from investing in the stock market. Interestingly, it was also found that investors are not negatively influenced by the failures of people around them mainly because people share success stories more than failures.

v) Expert Influence

Investors are also influenced by stock market experts. Analysts many times have incentives to tout particular shares or are influenced by other analysts or previously released information. This leads to herding and inferior returns in the market.

29.6.2.3 Emotional Influences on Investors

i) Feelings of Regret

Fear of regret deals with the emotional reaction that people experience after realizing that they have made a mistake or error in making a decision. When an investment goes bad, people act irrationally by trying to avoid selling it to avoid feelings of regret. Sometimes investors avoid the possibility of regret by following what other investors are doing. They feel less embarrassed about losing money in popular investments than in unpopular investments. The theory of Regret can be explained in terms of cognitive dissonance which we have already discussed under 'Cognitive influences on investor Behaviour'.

ii) Loss Aversion

Investors find losses to be disproportionately painful as compared to the pleasure from the gains. Loss aversion makes people prefer things as they are. People will overvalue things that they already own leading to divergences between willingness to pay and willingness to accept. People are willing to pay less for an object that they do not yet own than they will accept when selling the same object that they already own. For example: If you own an antique art piece worth a market price of Rs. 50,000, you will not be willing to pay Rs. 50,000 for it at this point but would expect a much higher price for it if you sell it.

iii) Sunk Cost Fallacy

The sunk cost fallacy describes an emotional tendency to invest more money, time, and effort into a project where we have already invested even if a person knows that the investment is going to fail.

The sunk cost fallacy is closely linked to 'loss aversion'. Investors become emotionally involved in a stock or project that they are not willing to accept the failure. Thus, they invest more money to make that investment work. People have difficulty in letting go of the sunk cost and they make additional mistakes like:

- **Aggressive Investing:** People who were risk averse initially become more aggressive in investing after losing a certain amount of money. They start taking undue risks which might lead to huge losses.
- **Averaging:** People invest more in the stocks where they have lost money as they try to average out returns in those stocks. This may lead to even more losses.

1) Explain various cognitive influences on an investor.

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2) What is the sunk cost fallacy?

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3) Consider the following statements. Name the bias/effect that an investor suffers from.

- a) An investor believes that others are better informed and thus follows them _____
- b) An investor attributes success to his ability and failures to external factors. _____
- c) An investor sells assets that have made financial gains while holding on to assets that are losing money. _____
- d) An investor finds losses to be disproportionately painful as compared to the pleasure from the gains. _____

29.7 LET US SUM UP

Traditional finance has been based on the concept of “efficient markets” which states that the price of an asset or security is ‘right’ as it reflects its fundamental value. Financial theory was governed by the idea that investors are rational.

However, over time, it was felt that the traditional theory was not able to explain market volatility and unpredictability in the market, market crashes, irrational investor Behaviour, stock market bubbles etc. These were explained by an alternative approach termed ‘Behavioural Finance’ which focuses on cognitive, sociological and psychological influences on the decision-making of investors and financial markets. More than contrasting, behavioural finance provides an alternative approach to traditional finance to have a better understanding of the market and investors. Behavioural finance draws from real life that investors are sometimes irrational, and biased and are influenced

by various social and psychological factors. It evolved mainly in the 1980s when Behavioural finance evolved with an attempt to explain stock market bubbles and other market anomalies.

In this unit, we also discussed about efficient markets hypothesis and its anomalies. According to EMH, markets are rational and prices of stocks fully reflect all available information and since the information is readily available, the price of securities quickly adjusts to the change. There are fundamental, technical and Calendar or season and all anomalies. These anomalies reflect occurrences that deviate from the predictions of the efficient markets hypothesis and undermine its core assumptions. Behavioural Finance also explains various market puzzles. The equity premium puzzle refers to the empirical fact that stocks have greatly outperformed bonds over the last century. Stocks have earned disproportionately higher returns than bonds, however, people still buy bonds. Behavioural finance explains this puzzle through the concepts of myopic loss aversion and Ambiguity aversion. The volatility puzzle reflects the high variability of stock prices having large dispersions. Behavioural finance attempts to solve this puzzle through various cognitive, social and emotional biases that investors have. This puzzle states that stock returns are predictable from price-dividend ratios. Behavioural finance is explained in terms of behavioural biases like the disposition effect, momentum effect and post-earning announcement drift.

This unit also discussed about four prominent heuristics that people follow while making decisions. Heuristics or ‘rules of thumb’ are imperfect but functional shortcuts that help investors simplify complex situations and make quick decisions. These heuristics are The anchoring and adjustment heuristic (people employ a certain starting point (“the anchor”) and make adjustments until they reach an acceptable value over time), the Representativeness heuristic (In situations of uncertainty, people evaluate the probability of an uncertain event by comparing it to an existing prototype that already exists in our minds), Availability Heuristics (assess the probability of occurrence of the event based on the ease with which it comes to the mind), Affect Heuristic (helps an individual to assign probabilities of occurrence of an event based on how a person feel about it).

We also discussed cognitive influences (Over-Confidence, Cognitive Dissonance, Under-reaction and Over-reaction, Disposition effects, Diversification bias, Mental Accounting), social influences (Herding, Reputation, Beauty contests, Success stories and advice from family, expert influence), and emotional influences (Feelings of Regret, Loss aversion, Sunk cost fallacy) on an investor.

29.8 KEY WORDS

Affect Heuristic : Affect heuristic helps individuals to make decisions based on how a person feels about it.

Ambiguity aversion: Ambiguity aversion is a preference for known over unknown risks by an individual.